LESSON 3.5

SOLVING REAL-WORLD PROBLEMS WITH RATIONAL NUMBERS – PART 1

LESSON INTRODUCTION

MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.

LEARNING TARGETS

- I can solve problems with one or two operations by adding and/or subtracting rational numbers.

BENCHMARK COHERENCE

BUILDING ON

MA.6.NSO.2.3 Solve multi-step, real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.

WORKING TOWARD

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

ESSENTIAL QUESTIONS

- How do you solve multi-step problems involving addition and subtraction of rational numbers?

LESSON NARRATIVE

Lesson 5 opens a series of lessons on solving problems using any of the four operations in a variety of real-world contexts. Students focus exclusively on addition and subtraction in real-world contexts in this lesson before transitioning to multiplication and division in Lesson 6 and extending to more complex scenarios in Lessons 7 and 8.

COMPONENT SUMMARY

3.5.1 WARM-UP *(~5 MIN.)*

Identify patterns in division equations

3.5.2 GUIDED INSTRUCTION *(10-15 MIN.)*

Complete tables to add and subtract rational numbers

3.5.3 COLLABORATION *(10-15 MIN.)*

Determine which account has the greatest value after several transactions

3.5.4 GUIDED PRACTICE *(5-10 MIN.)*

Determine change in and final elevation

3.5.5 YOUR TURN! *(5-10 MIN.)*

Apply rules for adding and subtracting rational numbers to real-world contexts

3.5.6 WRAP-UP *(~5 MIN.)*

Sometimes, always, or never growth mindset reflection

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Credit
- Debit
- Equations
- Expressions

MATERIALS

- (Optional) Chart paper
- (Optional) Markers

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may have trouble representing values in the questions with expressions and equations.
- Students may struggle to add and subtract mixed numbers with different denominators.
- Students may struggle to find the least common multiple when adding and subtracting mixed numbers.

THINGS TO CONSIDER

- The lesson concentrates on adding and subtracting rational numbers.
- Consider your students' experience with banks and checking accounts. If needed, discuss what students know about how bank accounts work, and have them read through the question to note any language they are unfamiliar with.
- If time is an issue, then consider assigning Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

3.5.1 Warm-Up provides an opportunity for students to engage in a Notice and Wonder routine. Encourage students to wonder why the quotient is the same in equation despite the changing numbers. This will position students to begin considering how the decimal point impacts the quotient and why that might be.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

This lesson extends on previous student learning by applying addition and subtraction of rational numbers to real-world contexts. Encourage students to connect the mathematical content of the lesson to their own lives through the examples provided.

DIFFERENTIATE INSTRUCTION

Consider utilizing the On-Ramp to 7th Grade Accelerated Math Learning Tool to provide scaffolding for students in need of additional support, particularly students who learn well with Study Expert videos and practice (available in English and Spanish). Videos provide additional examples and practice for real-world addition and subtraction, as well as a variety of both fraction and decimal operations for different portions of the lesson. Students can engage with On-Ramp videos on the following topics for additional support:

- Adding Fractions with Unlike Denominators
- Subtract Fractions with Unlike Denominators (3.5.2 Guided Instruction question 2)

3.5.1 WARM-UP: NOTICE AND WONDER

TEACHER MOVES

Notice and Wonder

Notice and Wonder activities allow students to lead the conversation. Encourage students to notice that the decimal place is changing in each equation while the quotient remains the same. Highlight student responses that recognize this and wonder what that means for the missing quotients.

3.5.1 Warm-Up: Notice and Wonder

Several division equations are shown.

$0.28 \div 0.07 = \boxed{}$

$2.8 \div 0.7 = \boxed{}$

$28 \div 7 = 4$

$280 \div 70 = 4$

$2{,}800 \div 700 = 4$

$28{,}000 \div 7{,}000 = 4$

1. What do you wonder about the pattern?

   Answers will vary. Will the pattern continue with the first two expressions? How does the decimal point affect the solution?

2. What do you notice about the equations?

   Answers will vary. The solution to all the expressions in the pattern is 4.

3. Use the pattern in the equations to determine the quotients of the first two equations.
   $0.28 \div 0.07 = 4$
   $2.8 \div 0.7 = 4$

3.5.2 GUIDED INSTRUCTION: ADDING AND SUBTRACTING RATIONAL NUMBERS

TEACHER MOVES

Students may decide the service fee is subtracting the positive number (the fee), while other students may decide to add the fee as a negative value. Both ways work, but make sure they understand not to both subtract and use a negative number to represent the fee. The number of rows given on the page is not an indication of the answer. There may be empty rows.

DIFFERENTIATE INSTRUCTION

The tables provide a visual means for students to organize the information presented in the question, providing a visual entry point for learners. Encourage students to discuss their strategies for solving with a partner to provide additional auditory entry points.

3.5.2 Guided Instruction: Adding and Subtracting Rational Numbers

1. Alexis has a monthly subscription to the video streaming service ViewNow for $7.20 per month. For her birthday she received a ViewNow gift card in the amount of $55.00.

   a. Write an expression using addition that represents the balance on the card after one month if the only charge on the gift card is the monthly subscription.
   $55 + (-7.20)$

   b. Complete the table by filling in values on each blank line to record the activity on the account each month. Continue each month to determine how many months it will take before Alexis does not have enough money to pay the monthly fee using the gift card. Part of Month 1 has been completed for you.

Month	Starting Balance	Change in Account	Ending Balance
1	\$55.00	\$55.00 + (−\$7.20)	\$47.80
2	\$47.80	\$47.80 + (−\$7.20)	\$40.60
3	\$40.60	\$40.60 + (−\$7.20)	\$33.40
4	\$33.40	\$33.40 + (−\$7.20)	\$26.20
5	\$26.20	\$26.20 + (−\$7.20)	\$19.00
6	\$19	\$19.00 + (−\$7.20)	\$11.80
7	\$11.80	\$11.80 + (−\$7.20)	\$4.60
8			
9			

   c. How many months will it take before Alexis does not have enough money left on the gift card to pay the monthly charge?
   7 Months

2. The **departure from the average** is the difference between the actual amount of rain and the average amount of rain for a given month. The departure from the average rainfall, in inches (in.), for Orlando, Florida for June, July, and August is shown in the table.

June	July	August
$-3\frac{19}{20}$ in.	$-1\frac{7}{10}$ in.	$4\frac{5}{8}$ in.

a. If the average rainfall for June is $7\frac{29}{50}$ in., what was the actual rainfall in June?

$3\frac{63}{100}$

b. What is the difference in the departure from the average rainfall from July to August?

$4\frac{5}{8}-\left(-1\frac{7}{10}\right)=6\frac{13}{40}$

3.5.2 GUIDED INSTRUCTION: ADDING AND SUBTRACTING RATIONAL NUMBERS (CONTINUED)

TEACHER MOVES

Departure from the average may be a new term for students but they have likely used the idea before when comparing values to averages. Pause to have students consider contexts where they may have used the idea themselves. One possible context is grades. If a student makes an 85 on a test but their test average is 91 then the 85 is a difference of 6 from the average. An example that students can relate to may be helpful before completing questions 2a and 2b.

Students may struggle to find the departure from the average. Ask guiding questions to help students subtract mixed numbers with different denominators:

- What is the least common multiple for these denominators? How can you use LCM to find the departure from the average?
- What operation can you use to find the departure from the average? Why does that work?

3.5.3 COLLABORATION: ACCOUNT TRANSACTIONS

TEACHER MOVES

Before releasing students read the introduction and ask students to give which values are debit in Account 1 and which values are credit in Account 1. Encourage students to use mental math and estimation to answer question 1.

DIFFERENTIATE INSTRUCTION

Have each group write down the steps they used to solve the question with words and with math notation on a poster board. Groups then share their poster board with the class and explain their steps. Allow other classmates to ask questions to foster mathematical discussion.

3.5.3 Collaboration: Account Transactions

The table shows transactions for four different checking accounts. A negative number represents a debit, or removing money from the account. A positive number represents a credit, or adding money to the account.

Account 1	Account 2	Account 3	Account 4
−$38.50	$250.00	−$14.00	−$86.80
$126.30	−$135.20	$99.90	−$570.00
$429.40	$35.50	−$82.70	$100.00
−$265.00	−$62.30	−$1.50	−$280.10

Work with your partner to complete the following using the information from the table.

1. Each checking account had the same starting balance. Without performing any calculations, discuss with your partner which account will have the most money in it after completing the transactions shown. Summarize your discussion.

   Answers will vary. Account 1 will have the most money after completing the four transactions. Both accounts 3 and 4 have 3 debits/negatives. Accounts 1 and 2 have only 2 debits, but it seems that Account 1's debits are smaller in relation to the credits in the account.

TEACHER MOVES

Students may struggle to understand that they are not to find the amount of money in the account but the change in the balance. Students may wonder what the beginning balance is. Remind them that all accounts had the same beginning balance.

Ask questions to guide students who are struggling:

- What do you notice about the positive amounts in each account? Why account has the most credits?
- How can you use what you know about positive and negative integers to predict which account will have the most money?

TEACHER MOVES

Encourage students to discuss the strategies each group used to find the overall change in balance for each account. Draw attention to groups whose strategies are particularly efficient and engage students in discussion about why particular strategies might be better to use than others.

2. With your partner, decide who will calculate the overall change in the account balance for Accounts 1 and 2, and who will calculate them for Accounts 3 and 4. Then, find the overall change in balance for each account after applying the transactions.

Account 1 Balance	Account 2 Balance	Account 3 Balance	Account 4 Balance
$252.20	$88	$1.70	−$836.90

3. Discuss with your partner which strategies you used for finding the overall change in balance for each account. Then, check your work and make any revisions if necessary.

   Answers will vary. The strategy that I used to find the ending balance was adding the credits and then subtracting the debits.

4. Which account will have the greatest final balance?

   Account 1 will have the greatest final balance.

5. Which account will have the smallest final balance?

   Account 4 will have the smallest final balance.

3.5.4 Guided Practice: Change in Elevation

1. Sasha spends fall break backpacking through the beginning part of the Badwater Basin Trail in Death Valley. Her journey includes climbing up and down the hills and mountains. Sasha's changes in elevation between each checkpoint of her trip are shown in the table.

   a. Complete the table to determine Sasha's change in elevation for each checkpoint. The first row has been completed to show that the change in elevation is determined by subtracting the beginning elevation from the final elevation.

Check-point	Beginning Elevation (ft)	Final Elevation (ft)	Expression	Change
1	−266.5	−279	−279 − (−266.5)	−12.5
2	−279	−282	−282 − (−279)	−3
3	−282	−240.3	−240.3 − (−282)	41.7
4	−240.3	280.1	280.1 − (−240.3)	520.4
5	280.1	755.2	755.2 − 280.1	475.1
6	755.2	1191.7	1191.7 − 755.2	436.5

   b. Which checkpoint had the greatest change in elevation?
   Checkpoint 4

   c. What do the signs of the change in elevation mean in this context?
   Answers will vary. A negative change represents a decrease in Sasha's elevation and a positive change represents an increase in Sasha's elevation.

3.5.4 GUIDED PRACTICE: CHANGE IN ELEVATION

TEACHER MOVES

Consider providing students with some background on Death Valley. Students in Florida often have little experience with elevation change unless they leave the state. Showing students pictures can help students gain an understanding that an elevation on land can have a negative value.

3.5.5 YOUR TURN!

TEACHER MOVES

The contexts may not be familiar to students. Encourage students to use the information in the context to determine what to do. Placing students in new situations helps them to apply the content they have learned. Allow students to have productive struggle. Direct students with questions that may help them retrieve the information themselves.

As you circulate note whether students are struggling with operations or with the context. Use this information to inform your instruction on the remaining lessons of this section.

3.5.5 Your Turn!

1. The table shows the change in value for a share of stock in a company at the end of each week during the month of December. A negative value represents the amount the share decreased, and a positive value represents the amount the share increased.

December
−\$22.65
\$225.30
−\$106.42
\$91.10

If the value of the stock began at \$75.37 at the start of the month, what is the final value of the share at the end of the month?

$\$75.37 - \$22.65 + \$225.30 - \$106.42 + \$91.10 = \$262.70$

2. Samuel is practicing for his town's annual high school quarterback football-throwing competition. Last year's champion threw the football $40\frac{1}{4}$ yards (yd). A summary of Samuel's throws in relation to the throws of last year's champion is shown. Complete the table to determine how far Samuel threw the football during each of his tries.

Throw	Comparison of Samuel's Throw to Last Year's Champion	Expression	Yards Thrown
1st	$+3\frac{1}{3}$ yards	$40\frac{1}{4} + 3\frac{1}{3}$	$43\frac{7}{12}$
2nd	$-\frac{9}{7}$ yards	$40\frac{1}{4} + \left(-\frac{9}{7}\right)$	$38\frac{27}{28}$
3rd	$-2\frac{6}{13}$ yards	$40\frac{1}{4} + \left(-2\frac{6}{13}\right)$	$37\frac{41}{52}$

3.5.6 Wrap-Up: Growth Mindset

1. As you reflect on today's lesson, circle the word *Sometimes*, *Always*, or *Never* to indicate how often each statement describes you.

   Answers will vary.

When I work hard on a problem, I expect that the answer will be correct.	Sometimes	Always	Never
There is no point in trying to do a problem if I know I won't get the correct answer.	Sometimes	Always	Never
Even if I made a mistake, I can still learn from working through a problem.	Sometimes	Always	Never
I can show that I understand how to solve a problem, even if there is an error in my work.	Sometimes	Always	Never
If my answers are incorrect even after working extremely hard, I still choose to work hard on the next problem.	Sometimes	Always	Never

3.5.6 WRAP-UP: GROWTH MINDSET

TEACHER MOVES

Consider prompting students to choose one of the statements and provide an example and justification for their response. Encourage them to use an example from this lesson or Unit 3.

Use student feedback to help you consider which students may need to be encouraged to embrace a growth mindset.